Characterizing Differences Between VR-Based and Paper-Based Handwriting Assessment: A Pilot Study

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***Abstract*—This paper investigates differences between VR-based and paper-based handwriting assessment using a combination of a VR headset and a digitizing tablet. Handwriting has progressively lost space in everyday practice with the widespread adoption of digital devices, and the long-term goal of this work is to enable VR-based handwriting instruction that improves motivation and engagement, particularly in children. As a necessary precondition, we investigate whether the proposed setup produces assessment outcomes comparable to conventional pen-and-paper administration. The same structured protocol was applied in both conditions, covering alphabet writing, pangram copying, figure filling,and graphomotor pattern reproduction. In a pilot study with eleven adults, writing speed in VR was on average 3.7 times slower than on paper, and letter malformation rates were consistently higher in VR(23.3%–45.7%) than on paper (0%–13.4%), with no overlap between the two ranges. Figure filling was the only task to show close agreement between modalities, with four participants obtaining identical scores in both conditions. These results characterize which aspects of handwriting assessment diverge between VR-based and paper-based administration,identify figure filling as a candidate for equivalence, and inform the design of a full equivalence study.**

***Index Terms*—virtual reality, handwriting assessment, measurement equivalence, digitizing tablet, graphomotor evaluation**

I. Introduction

Handwriting is a complex skill whose acquisition depends on the integrated development of fine motor function, visual perception, and cognitive processes [1], [2]. Once the cognitive-motor skill of writing is acquired, it supports knowledge construction and communication throughout life [3]. In recent years, however, the widespread adoption of digital devices among younger generations, often referred to as digital natives [4], has progressively displaced handwriting practice,with reviews of the literature reporting associated effects on children's early writing outcomes [13].

Virtual Reality (VR) emerges as a promising medium to address this gap, offering immersive environments that can re-engage learners with handwriting while retaining the cognitive benefits of the physical act of writing. Kern et al. [5] demonstrated the technical feasibility of handwriting in VR with consumer-grade hardware, showing that legible handwriting is achievable on physically aligned virtual surfaces. However, technical feasibility alone does not establish measurement equivalence: whether writing produced in a VR environment yields results equivalent to conventional pen-and-paper writing remains an open question.

To investigate this question, this paper presents a novel combination of a head-mounted display (HMD) and a pen digitizing tablet to replicate the physical act of writing on a real surface within an immersive virtual environment (Fig. 1A). Following the surface alignment principles of Kern et al. [5], our method renders digital ink on a virtually aligned canvas that precisely corresponds to the physical tablet area, preserving the passive haptic feedback of conventional writing. We adopt an adaptation of the handwriting assessment protocol proposed by Cardoso [6] — covering alphabet writing, pangram copying, figure filling, and graphomotor pattern reproduction — with adjustments to task duration and text length to fit a single experimental session, and automatically evaluate responses, enabling direct metric comparison between physical and VR conditions.

A pilot study with eleven adults (N=11) reveals systematic differences between the two conditions: writing speed in VR was on average 3.7 times slower than on paper, and letter malformation rates were consistently higher in VR (23.3%–45.7%) than on paper (0%–13.4%), with no overlap between the two ranges. Figure filling was the only task to show close agreement between modalities, with four participants obtaining identical scores in both conditions. These results characterize which aspects of handwriting assessment transfer well to VR and which do not,laying the groundwork for a full equivalence study and, eventually, a richer virtual environment aimed at increasing engagement with handwriting practice (Fig. 1B).

II. Related Work

This section reviews work directly relevant to the three pillars underlying our research: handwriting performance across writing surfaces, pen-based interaction in VR, and digital handwriting assessment validation.

A. Handwriting Performance Across Writing Surfaces

Gerth et al. [8] compared adult handwriting on paper and on a digitizing tablet, finding that participants wrote faster on the tablet due to its lower friction but adapted their motor execution to compensate. A companion study [9]extended this comparison to preschoolers, second graders, and adults,finding

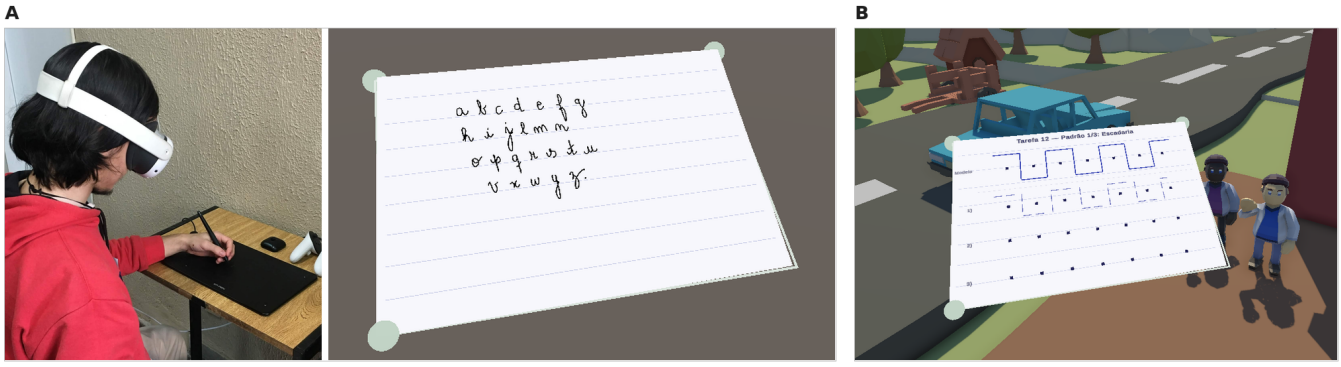


Fig. 1. Application overview. (A) Current setup: participant writing on the digitizing tablet while wearing the HMD, alongside the participant view showing the alphabet writing task (T01) on the virtual canvas. (B) Planned gamified version of the application, incorporating a 3D environment, characters, and contextualized task presentation as a future direction.

that the smoother tablet surface increased writing velocity and posed additional motor control demands for less automated writers, an effect further shown by Alamargot et al. [10] to be modulated by writing expertise. Together, these studies establish that the writing surface influences handwriting dynamics even in skilled adults, motivating the question of whether a VR environment —introducing perceptual differences beyond surface friction — produces larger or more systematic effects on handwriting performance.

B. Pen-Based Interaction in VR

Kern et al. [5] introduced the Off-The-Shelf Stylus (OTSS) framework for handwriting and sketching on physically aligned virtual surfaces using consumer-grade XR controllers paired with a digitizing tablet. Their reference implementation achieved a mean stylus precision of 0.98 mm, a surface accuracy of 0.60 mm, and an average digital ink latency of 79.40 ms. A usability study with ten participants reported low task load, high usability, and legible handwriting output across visuomotor and graphomotor tasks structurally similar to those of established paper-based protocols. OTSS establishes that legible handwriting is achievable in VR with consumer hardware, and explicitly identifies physical surface alignment as a requirement for preserving the passive haptic feedback characteristic of conventional writing — the same design principle adopted in this work. More recently, Fourrier et al. [14] investigated handwriting as a text-entry technique in industrial VR applications,examining the influence of board orientation and sensory feedback on performance; their work reinforces that handwriting in VR remains sensitive to setup-specific factors even when technically feasible.

C. Digital Handwriting Assessment Validation

Guzmán and Grajo [11] developed and validated TeleWrite, a telehealth-based handwriting assessment administered remotely via digital tools. Using Rasch analysis, the authors evaluated construct validity and reliability of the digital instrument, demonstrating that structured handwriting assessments can be administered and scored outside the traditional paper-based clinical setting while retaining psychometric validity. This work illustrates current approaches to digital handwriting assessment validation and positions the equivalence question addressed in our study within a broader clinical and educational context.

While these works confirm that surface properties affect handwriting dynamics and that legible handwriting is achievable in VR, none of them directly compares VR-based and paper-based administration of the same standardized assessment protocol. We address that gap by implementing the adapted TPEM battery in both conditions and measuring performance equivalence across tasks. More broadly, the SVR conference has a long history of studies applying VR to motor assessment and rehabilitation tasks [16], though direct comparisons against standardized paper-based batteries, as proposed here, remain uncommon in this body of work.

III. Theoretical Background

This section establishes the theoretical foundations underlying the design and evaluation of this work. It covers three main axes: the nature of handwriting as a complex skill and its assessment (III-A), the similarity metric used for automated stroke evaluation (III-B), and the path from the technical feasibility of handwriting in VR to the open question of measurement equivalence (III-C).

A. Handwriting as a Complex Skill

Handwriting is a complex skill that requires the continuous integration of fine motor function, visual perception, visuomotor coordination, and cognitive processes [6]. At the neuromotor level, producing written letters demands upper limb coordination characterized by a continuous flow of movement, requiring writers to store letter images in memory, transcribe them into motor commands, and adjust execution according to visual feedback [6]. When this process becomes automatic (typically around age 8 to 9) handwriting no longer occupies attentional resources, freeing working memory for higher-order tasks such as text composition [6]. Alternative theoretical

frameworks, such as the writing formulation hypothesis, offer a complementary account of how this competence develops in children [15]. When automatization does not develop adequately, studies estimate that between 5% and 30% of school-age children may present handwriting difficulties, with consequences for legibility, writing speed, and academic performance [6].

Assessing handwriting rigorously requires both a structured task set and well-defined metrics. The tasks implemented in this work are drawn from the Handwriting Proficiency Test (TPEM — Teste de Proficiência da Escrita Manual) developed by Cardoso [6], which covers alphabet writing from memory, pangram copying, graphomotor pattern tracing, and geometric form reproduction. The original protocol organizes these tasks into named batteries (e.g., BAT-EM for handwriting-specific tasks and BAT-PVM for perceptual-visuomotor tasks); for clarity, this paper refers to the entire adapted battery collectively as TPEM. These tasks engage different cognitive, linguistic, and motor processes and together provide a broad basis for characterizing handwriting performance across its visuo-motor and graphomotor dimensions. Handwriting dynamics (including writing velocity, number of pen lifts, and in-air time) are recognized as objective indicators of motor automaticity and writing quality [8], and provide the measurement basis for comparing performance across conditions.

B. Stroke Similarity Metric

Automated evaluation of handwriting strokes requires a metric capable of quantifying the geometric similarity between two point sets regardless of their size, sampling density, or traversal direction. In this work, stroke similarity is computed using a bidirectional mean nearest-point distance, a practical approximation of the Wasserstein distance [7] for discrete point sets. The Wasserstein distance, also known as the Earth Mover's Distance, quantifies the minimum cost required to transform one point distribution into another [7].

Given a captured stroke represented as a normalized point set $P = \{p_i\}$ and a reference model $Q = \{q_j\}$, both normalized to a unit bounding box, the distance is computed as:

$$d(P, Q) = \max\left(\frac{1}{|P|} \sum_{p \in P} \min_{q \in Q} \|p - q\|,\ \frac{1}{|Q|} \sum_{q \in Q} \min_{p \in P} \|q - p\| \right) \tag{1}$$

For each point in P, the nearest point in Q is found and the mean of those distances is computed, and vice versa. The final distance is the maximum of the two means, penalizing gaps in both directions and making the metric robust to differences in stroke direction and starting point.

The resulting distance is converted to a similarity score in the range $[0, 100]$ via an exponential decay function:

$$s = 100 \cdot e^{-3d} \tag{2}$$

where higher values of s indicate closer correspondence to the expected form. This formulation was calibrated for handwriting strokes captured on a digitizing tablet, where reference values include $d = 0.08 \rightarrow s \approx 79$ (good correspondence) and $d = 0.30 \rightarrow s \approx 41$ (weak correspondence).

C. Handwriting in VR: From Feasibility to Equivalence

The technical feasibility of handwriting in VR has been established by prior work on stylus-based interaction in immersive environments. Kern et al. [5] showed that handwriting on physically aligned virtual surfaces is legible and usable, identifying surface alignment as a key requirement: the rigid surface provides passive haptic feedback that preserves the natural hand-surface relationship of conventional writing. This work follows the same design principles, combining a head-mounted display (HMD) with a pen digitizing tablet whose surface is calibrated and virtually aligned so that digital ink corresponds precisely to the participant's physical writing area, enabling direct comparison of handwriting metrics collected in VR and on paper under identical task conditions.

However, confirming that handwriting is possible in VR is not the same as confirming that it produces equivalent results. Gerth et al. [8], [9] showed that merely changing the writing surface from paper to a digitizing tablet alters writing dynamics — velocity, pen lifts, and in-air time — even among experienced writers. If a change in surface friction is sufficient to affect motor execution, the perceptual differences introduced by a VR environment (including visual latency and the absence of direct visual contact with the hand) could produce larger and more systematic effects. Establishing whether VR-based and paper-based assessment produce equivalent results is therefore a necessary precondition for the responsible use of VR tools in handwriting assessment, and is the central empirical question motivating this work.

IV. Methodology

This section describes the method adopted to investigate measurement equivalence between VR-based and paper-based handwriting assessment. As shown in Fig. 2, after a surface calibration step the participant performs the same four tasks in both conditions, each followed by its corresponding evaluation, and the resulting scores feed the equivalence analysis. The four tasks — labeled T01 through T04 in this work — correspond to tasks T01, T04, T11, and T12 of the original protocol by Cardoso [6].

A. Data Collection Setup

Two hardware components support the digital condition: a pen digitizing tablet and a head-mounted display (HMD). The participant writes directly on the tablet surface using the accompanying digital pen, which captures stylus position and contact pressure continuously during each task. A configurable pressure threshold distinguishes active pen contact from hover, ensuring that only intentional strokes are recorded. The rigid tablet surface preserves the hand-surface relationship characteristic of conventional handwriting, providing passive haptic feedback throughout the session.

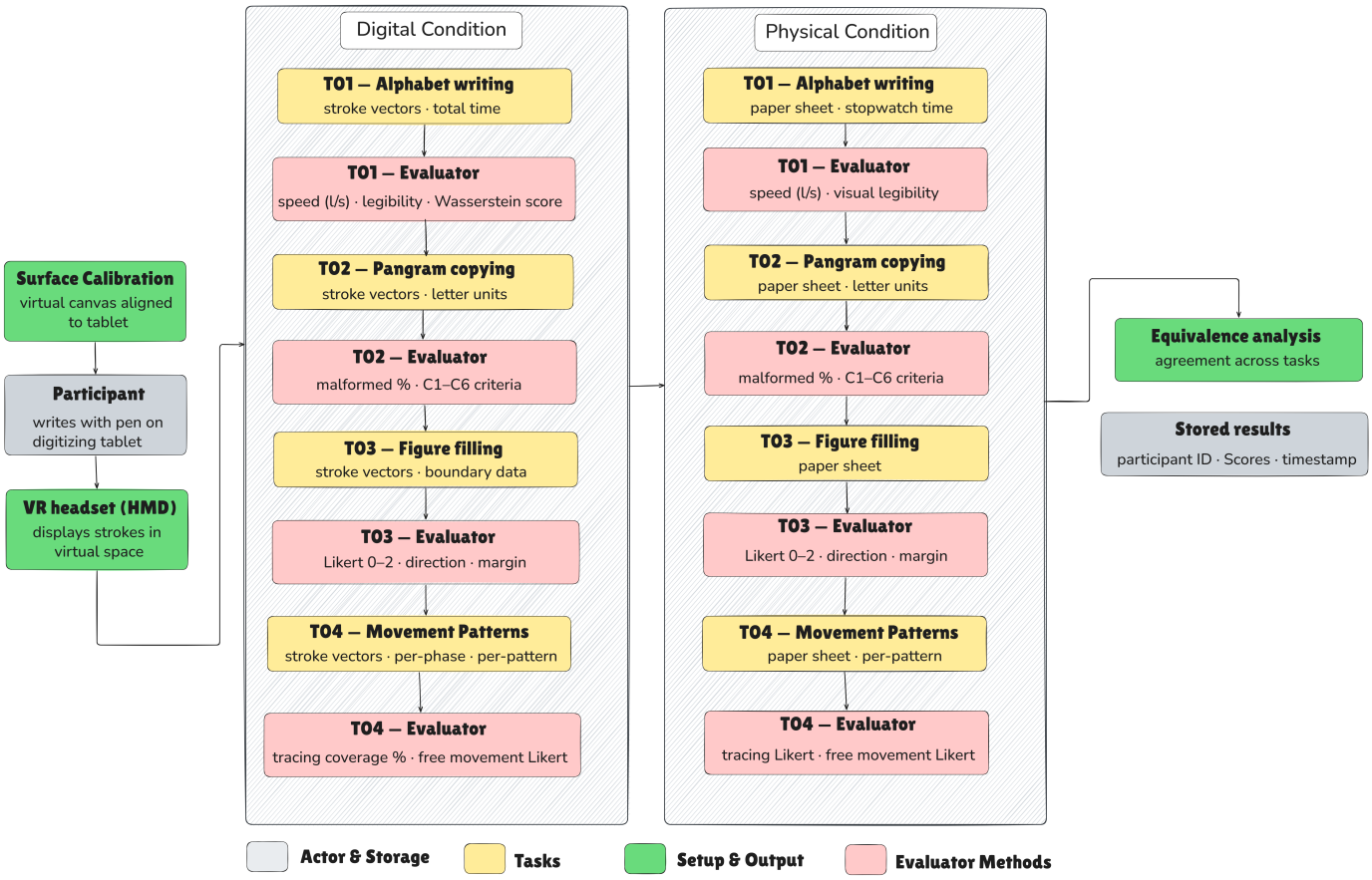


Fig. 2. Methodological pipeline of the proposed application. After surface calibration, the participant performs the four tasks in the digital condition (left) and the physical condition (right); each task is followed by its evaluation step. Scores from both conditions are stored per participant and compared in the equivalence analysis.

At the start of each session, a four-corner calibration procedure is performed: the participant marks each corner of the active writing area with the stylus, and the application computes the position, orientation, and scale of a virtual canvas that precisely overlaps the physical tablet surface. This alignment ensures that digital ink, task stimuli, and the physical writing area are co-located in the participant's visual field, preserving the spatial correspondence between motor execution and visual feedback.

Task administration was controlled exclusively via keyboard by the researcher, who managed all task transitions, phase advances, and session resets without direct interaction with the writing surface. This keyboard-only design prevents accidental interruptions during data collection, which is particularly relevant given that the digitizing tablet stylus is recognized by the operating system as a pointing device.

B. Digital Condition

In the digital condition, each stroke produced by the participant is recorded as an ordered sequence of two-dimensional points. Strokes are accumulated separately per task (and, in T04, per pattern and per phase) so that each evaluation step receives only the strokes belonging to the unit under analysis. At task completion, the corresponding evaluator processes the accumulated strokes as follows.

T01 — Alphabet writing (T01 in [6]). The writing area is divided into 26 zones, one per letter, and the strokes captured in each zone are grouped as that letter's production. Evaluation proceeds in three steps. First, each letter's strokes are smoothed to reduce sensor noise while preserving stroke boundaries. Second, the smoothed strokes are compared against the cursive reference model of the corresponding letter, extracted from the PRALER atlas [12] (Fig. 4): both the captured strokes and the reference model are normalized to a unit bounding box and compared using the bidirectional nearest-point approximation of the Wasserstein distance described in Section III (Eq. (1)), producing a similarity score in the range $[0, 100]$ (Eq. (2)). Third, each letter is classified from its score as legible ($s \geq 60$), partially legible ($30 \leq s < 60$), or illegible ($s < 30$); empty zones are counted as not written. Writing speed is computed as the number of legible letters divided by the total task duration, following [6].

T02 — Pangram copying (T04 in [6]). Captured strokes are grouped into letter units based on spatial proximity, and each unit is checked against six topographic criteria (C1–C6) derived from [6]: C1, abnormal size variation relative

to adjacent letters; C2, collision or overlap with adjacent letters; C3, presence of erasures or overwriting; C4, print form used instead of cursive; C5, incorrect use of uppercase; C6, significant deviation from expected letter form, measured against the PRALER reference models. A unit is classified as malformed if any criterion is met, and the task result is the percentage of malformed units relative to the total copied. The pangram used in this study comprises 98 letters covering all 26 letters of the alphabet, a reduced adaptation of the original text proposed by [6] to fit the time constraints of a single session.

T03 — Figure filling (T11 in [6]). For each of the three figures (circle, square, hexagon), the evaluator analyzes the stroke trajectories produced inside the figure region, verifying two aspects: whether the movement direction matches the instructed pattern (circular, vertical, or horizontal, respectively), and whether the strokes remain within the figure boundary. Each figure receives a Likert score from 0 to 2: 0 for incorrect movement direction or complete boundary violation; 1 for correct direction with partial boundary violation, or boundary respected with inconsistent direction; 2 for correct direction maintained within the boundary. The total score ranges from 0 to 6.

T04 — Three basic movement patterns (T12 in [6]). Each pattern (staircase, zigzag, loop) has two phases, evaluated separately. In the tracing phase, the dashed template shown to the participant is decomposed into its individual segments, and a segment is counted as covered when the participant's strokes pass sufficiently close to most of its points; the result is reported as the number of covered segments over the total (maximum 42 across the three patterns), expressed as a percentage. In the free reproduction phase, each repetition is first checked for pen lifts — continuous patterns reproduced with multiple disconnected strokes receive score 0 — and then compared against the reference pattern using the same shape similarity metric of Eq. (1), combined with a proportion check between the produced and expected pattern dimensions. Each repetition receives a Likert score from 0 to 2 (0: pattern not produced or pen lifted; 1: correct movement with incorrect proportions; 2: correct movement and proportions), for a total free movement score from 0 to 12.

C. Physical Condition

In the physical condition, participants completed the same four tasks on standardized protocol sheets (Fig. 3) using a conventional pen, and all evaluation was performed manually by there searcher after data collection, applying the same criteria used in the digital condition.In T01, the researcher timed the task with a stopwatch; writing speed was computed as the number of legible letters divided by the recorded duration,and each letter was classified visually as legible, partially legible, or illegible using the PRALER models as visual reference. In T02, each letter unit of the copied pangram was inspected against the same six topographic criteria (C1–C6), and the percentage of malformed units was computed over the total copied. In T03, each figure was scored visually on the same 0–2 Likert rubric, assessing movement direction and boundary compliance. In T04, the tracing phase was scored on a 0–2 Likert scale per pattern based on visual fidelity to the dashed template, and the free reproduction phase was scored with the same 0–2 rubric per repetition applied in the digital condition.

D. Output and Comparison

At the end of each task, the computed metrics were stored in a structured record identified by participant code and timestamp, containing the task identifier, the metric values, and per-item detail where applicable — letter-level results for T01 and T02, figure scores for T03, and per-pattern scores for T04. These records constitute the primary data source for the equivalence analysis reported in Section VI, where digital and physical scores are compared directly across all tasks and participants.

V. Experiment

This section describes the experimental procedure adopted for both conditions of the adapted protocol, including participant recruitment and session structure. The evaluation criteria applied in each condition are described in Section IV.

A. Participants

Eleven adults (N=11) participated in the study, identified with IDs IF-P01 through IF-P11. No inclusion or exclusion criteria were applied beyond the requirement that participants be 18 years of age or older. The study was conducted by the same researcher across all sessions, who also served as the protocol administrator.

B. Session Structure

Each participant completed both conditions — physical and VR — in a single session. To control for order effects, participants were counterbalanced across conditions: six participants began with the physical condition followed by the VR condition (IF-P02, IF-P03, IF-P05, IF-P07, IF-P09, IF-P11), and five participants began with the VR condition followed by the physical condition (IF-P01, IF-P04, IF-P06, IF-P08, IF-P10). In both conditions, tasks were administered in a fixed sequence: T01 $\rightarrow$ T02 $\rightarrow$ T03 $\rightarrow$ T04. No familiarization or training session was provided prior to either condition. In the VR condition, participants remained seated throughout the entire session.

C. Physical Condition

In the physical condition, participants completed the four tasks on standardized protocol sheets (Fig. 3) using a conventional pen. Task timing was recorded manually by the researcher using a stopwatch, and responses were evaluated manually after data collection, following the criteria described in Section IV-C. The PRALER cursive letter models (Fig. 4) served as visual reference for letter classification.

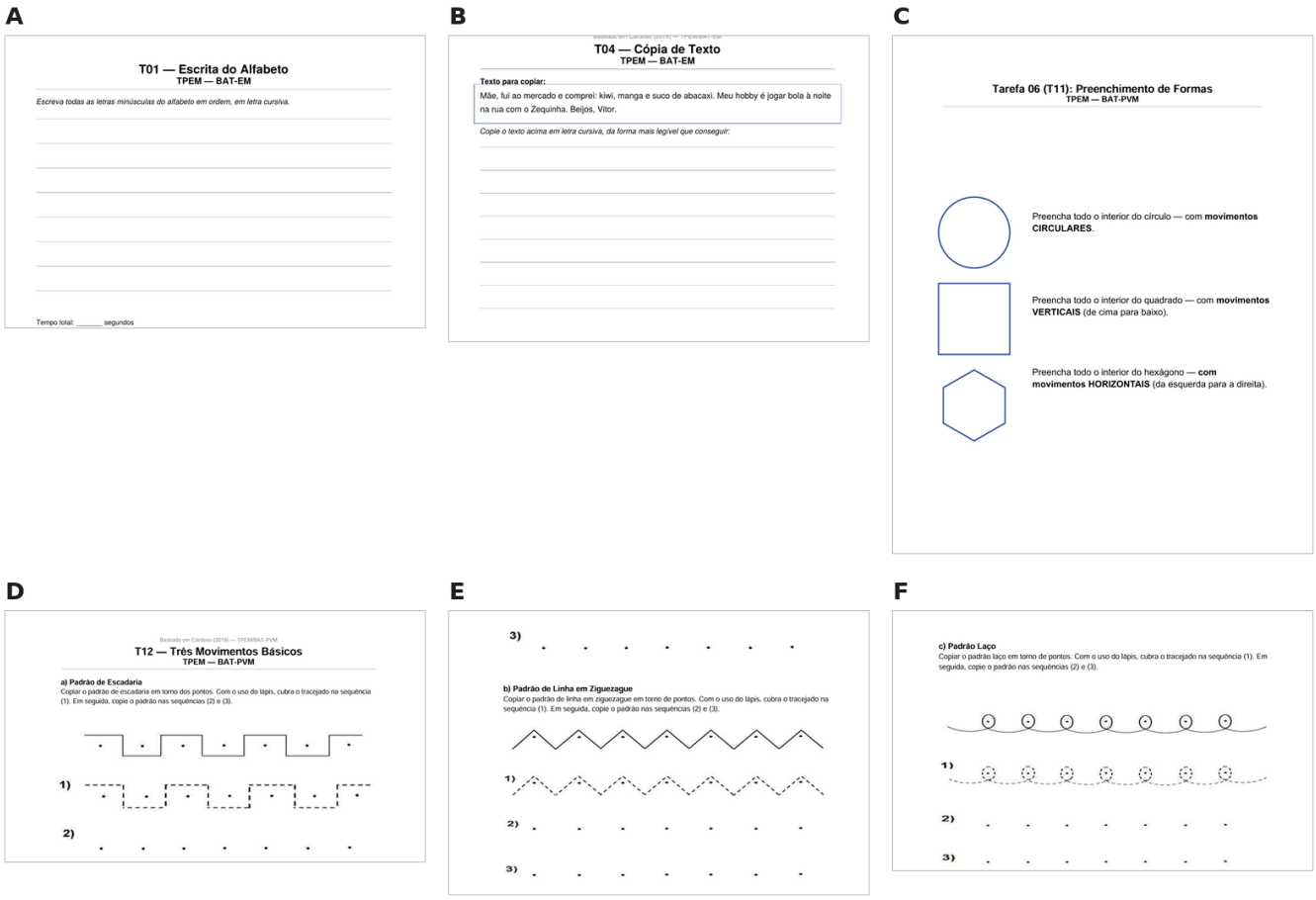


Fig. 3. Physical protocol sheets administered to participants, adapted from [6]. (A) T01 — Ruled lines for cursive alphabet writing (a–z), with total time recorded by the researcher. (B) T02 — Pangram text for cursive copying (T04 in the original protocol). (C) T03 — Figure filling (T11 in the original protocol): circle with circular movements, square with vertical movements (top to bottom), and hexagon with horizontal movements (left to right). (D) T04 (T12 in the original protocol), staircase pattern: solid model with guide dots, dashed tracing line (1), and free reproduction lines (2, 3). (E) T04, zigzag pattern: same structure as (D). (F) T04, loop pattern: solid model with circles at peaks, dashed tracing line (1), and free reproduction lines (2, 3).

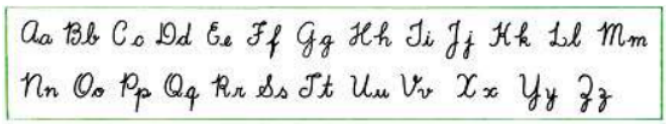


Fig. 4. Cursive letter reference models from the PRALER protocol [12], used as visual standard for letter classification in T01 and T02.

D. VR Condition

In the VR condition, participants used a digital pen on a digitizing tablet to perform the same four tasks. Handwriting strokes were captured in real time and rendered on a virtual canvas aligned to the physical tablet surface, displayed through the HMD. Task administration was controlled by the researcher through a separate monitor interface, with all task transitions managed via keyboard commands. Both conditions present the participant with the same task structure: Fig. 5 shows the staircase pattern of T04 as administered on the physical protocol sheet (A) and on the virtual canvas displayed through the HMD (B), illustrating the visual correspondence between the two conditions. Evaluation of VR responses was performed automatically by the application at the end of each task, following the criteria described in Section IV-B, storing the computed metrics per participant.

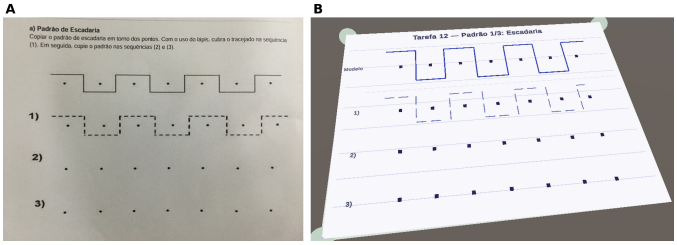


Fig. 5. The same task unit administered in both conditions, illustrated by the staircase pattern of T04. (A) Physical protocol sheet, with the solid model, guide dots, dashed tracing line (1), and free reproduction lines (2, 3). (B) Corresponding virtual canvas rendered in the participant's view, reproducing the same structure on the surface aligned to the digitizing tablet.

VI. Results

This section presents the results of a pilot equivalence study conducted with eleven participants (N=11) who completed both the physical and VR conditions of the adapted protocol. Results are reported by task, with direct comparison between conditions where data were valid in both. Tasks T01 and T02 provided the most direct metric comparisons, as the same evaluation criteria were applied in both conditions. Tasks T03 and T04 are also reported, with T04 presenting identified technical limitations in the VR condition that affected a subset of participants; these are discussed separately within the respective subsections. A supplementary video accompanies this submission, illustrating the application in use and the administration of the tasks in the VR condition.

A. T01 — Alphabet Writing

Task 01 was based on complete alphabet cursive handwriting from letter a to z in sequence using lowercase letters, evaluated in both physical and VR conditions. Two indicators were assessed: writing speed (letters per second) and legibility of the produced letters.

Writing speed was calculated by dividing the number of legible letters by the total timed duration from the beginning to the conclusion of the task, following the procedure adopted by [6]. Four participants (IF-P02, IF-P04, IF-P08, and IF-P10) did not complete the full alphabet in the VR condition and were therefore excluded from the speed comparative analysis. The remaining seven participants completed the task in both conditions.

Table I presents the writing speed results for participants with complete tasks. In the physical condition, speed ranged from 0.14 to 1.37 letters per second, while in the VR condition it ranged from 0.092 to 0.246 letters per second. All seven participants with complete tasks showed a reduction in writing speed in VR, with an average ratio of approximately 3.7× relative to the physical condition. IF-P03, whose physical speed (0.14 letters/second) was the slowest in the sample, showed the smallest ratio (1.5×), while IF-P11, the fastest writer on paper (1.37 letters/second), showed the largest (7.7×); the remaining five participants showed ratios between 2.5× and 4.1×. This pattern is consistent with findings from [8].

Legibility was calculated as the proportion of letters classified as legible or partially legible relative to the total number of written letters, following the classification criteria established by [6]. For the four participants with incomplete tasks, the calculation was performed over the letters effectively written. Table II presents the legibility results for all eleven participants.

In the physical condition, legibility ranged from 80.8% to 100%. In the VR condition, it ranged from 88.5% to 100%; however, every 100% VR value corresponds to one of the four participants who did not complete the alphabet, whose legibility was computed over fewer letters,which likely inflates this upper bound. Considering only the seven participants who completed the task in both conditions, physical legibility ranged from 80.8% to 100% (mean 95.6%) and VR legibility ranged from 88.5% to 96.2% (mean 92.3%) — i.e., the VR mean was slightly lower than the physical mean among completers. The four participants with incomplete tasks nonetheless achieved 100% legibility over the letters they did write, suggesting that the task interruption did not stem from difficulty producing legible handwriting.

TABLE I
T01 — Alphabet Writing Speed: comparison between physical and VR conditions.

Participant	Physical (letters/sec)	VR (letters/sec)
IF-P01	0.58	0.158
IF-P02	0.92	— *
IF-P03	0.14	0.092
IF-P04	1.08	— *
IF-P05	1.00	0.244
IF-P06	0.55	0.155
IF-P07	0.41	0.167
IF-P08	0.74	— *
IF-P09	0.65	0.246
IF-P10	1.13	— *
IF-P11	1.37	0.179

* Incomplete task in VR — excluded from comparative analysis.

TABLE II
T01 — Alphabet Writing Legibility: comparison between physical and VR conditions.

Participant	Physical (%)	VR (%)
IF-P01	96.15	92.30
IF-P02	88.46	— †
IF-P03	80.77	88.50
IF-P04	100.00	— †
IF-P05	100.00	96.00
IF-P06	100.00	88.50
IF-P07	92.31	96.20
IF-P08	96.15	— †
IF-P09	100.00	92.30
IF-P10	100.00	— †
IF-P11	100.00	92.30

† Incomplete task in VR — legibility computed over letters effectively written, not over the full alphabet (26 letters).

B. T02 — Pangram Copying

Task 02 consisted of copying a pangram in cursive handwriting, covering all 26 letters of the alphabet in a reduced text of 98 letters, adapted from the original protocol proposed by [6]. The evaluated indicator was the percentage of malformed letters relative to the total number of letters copied, following the same classification criteria applied in both conditions.

Table III represents the malformed letter results for all eleven participants. In the physical condition, the percentage of malformed letters ranged from 0% to 13.40%. In VR condition, the range was from 23.33% to 45.68%. Although the VR condition consistently showed higher percentages of malformed letters across all participants, the relative ordering between participants is partially preserved, suggesting that individual handwriting characteristics influence performance in both conditions. The systematically higher malformation

rate in VR is consistent with the speed reduction observed in T01, and may reflect the perceptual-motor adaptation demands of writing while viewing output through a head-mounted display, as noted by [5].

TABLE III
T02 — PANGRAM COPYING: PERCENTAGE OF MALFORMED LETTERS IN PHYSICAL AND VR CONDITIONS.

Participant	Physical (%)	VR (%)
IF-P01	2.04	30.16
IF-P02	1.02	27.66
IF-P03	13.40	45.68
IF-P04	2.04	25.58
IF-P05	0.00	27.77
IF-P06	0.00	23.40
IF-P07	12.24	23.33
IF-P08	3.06	28.07
IF-P09	0.00	25.00
IF-P10	0.00	27.91
IF-P11	0.00	35.59

C. T03 — Figure Filling

Task 03 consisted of filling three geometric figures (a circle, a square,and a hexagon) with specific movement patterns: circular, vertical, and horizontal, respectively, following the instructions of the BAT-PVM battery of the TPEM protocol [6]. Each figure was scored on a Likert scale from 0 to 2, where 0 indicates the figure was not filled, 1 indicates filling with correct movement pattern but exceeding the boundaries, and 2 indicates correct filling within the boundaries. The total score per participant ranged from 0 to 6.

Table IV presents the total Likert scores for all eleven participants in both conditions. In the physical condition, scores ranged from 3/6 to 6/6. In the VR condition, scores ranged from 3/6 to 6/6 as well. Notably, T03 is the task that shows the closest agreement between physical and VR conditions across the dataset, with four participants obtaining identical scores in both modalities (IF-P01, IF-P07, IF-P10, IF-P11), suggesting that the filling movement patterns are well captured by the VR system using the digitizing tablet.

TABLE IV
T03 — FIGURE FILLING: TOTAL LIKERT SCORE IN PHYSICAL AND VR CONDITIONS (MAX = 6).

Participant	Physical (/6)	VR (/6)
IF-P01	6	6
IF-P02	4	3
IF-P03	3	5
IF-P04	5	4
IF-P05	5	6
IF-P06	4	6
IF-P07	3	3
IF-P08	5	4
IF-P09	5	6
IF-P10	5	5
IF-P11	4	4

D. T04 — Graphomotor Patterns.

Task 04 consisted of reproducing three basic handwriting movement patterns (staircase, zigzag, and loop) each presented in three lines: line 1 with a dashed template to be traced, and lines 2 and 3 for free reproduction without guidance. Two indicators were assessed separately: tracing coverage for line 1, and free movement quality for lines 2 and 3.

Tracing coverage in VR was affected by a technical limitation in the stroke separation algorithm, which classifies strokes by vertical Y position on the tablet surface: for participants IF-P06 through IF-P11,staircase and zigzag strokes fell outside the expected Y threshold and were recorded as zero coverage, independent of actual performance. This artifact, combined with the scale mismatch described above, motivates excluding tracing from the present comparative analysis; it is addressed further in Section VII and identified as a target for correction in future versions of the protocol.

For the free movement indicator, both conditions used the same Likert scale (0–2 per repetition per pattern, maximum 12). Table V presents the free movement results for all eleven participants. In the physical condition, scores ranged from 8/12 to 12/12. In the VR condition, scores ranged from 0/12 to 10/12. Participants IF-P04 and IF-P10 recorded 0% in the VR condition due to a technical limitation in the pen-lift detection algorithm, which required more than two continuous strokes to validate a movement — a threshold that proved overly restrictive for some writing styles. Excluding these two cases, VR free movement scores ranged from 16.7% to 83.3%, with IF-P03 and IF-P11 achieving the closest agreement with their physical scores. These technical limitations are identified as correctable system constraints and do not invalidate the overall feasibility of the VR protocol for assessing basic handwriting movement patterns.

TABLE V
T04 — THREE BASIC MOVEMENTS, FREE MOVEMENT: PHYSICAL VS. VR LIKERT SCORE (%).

Participant	Physical (%)	VR (%)
IF-P01	100.0 (12/12)	16.7 (2/12)
IF-P02	100.0 (12/12)	50.0 (6/12)
IF-P03	91.7 (11/12)	83.3 (10/12)
IF-P04	75.0 (9/12)	0.0 (0/12) †
IF-P05	100.0 (12/12)	66.7 (8/12)
IF-P06	100.0 (12/12)	66.7 (8/12)
IF-P07	66.7 (8/12)	66.7 (8/12)
IF-P08	91.7 (11/12)	66.7 (8/12)
IF-P09	100.0 (12/12)	66.7 (8/12)
IF-P10	100.0 (12/12)	0.0 (0/12) †
IF-P11	100.0 (12/12)	83.3 (10/12)

† Score zeroed due to pen-lift detection threshold artifact (see Section VI).

VII. DISCUSSION

The results of this pilot study provide initial evidence regarding the feasibility and partial measurement equivalence of the proposed application for administering an adapted handwriting assessment protocol in a virtual reality environment.

TABLE VI
SUMMARY OF RESULTS ACROSS ALL TASKS AND CONDITIONS (PHYSICAL VS. VR).

Part.	T01 Speed (l/s)		T01 Legib. (%)		T02 Malf. (%)		T03 Score (/6)		T04 Free (%)	
	Phy	VR	Phy	VR	Phy	VR	Phy	VR	Phy	VR
IF-P01	0.58	0.158	96.2	92.3	2.0	30.2	6	6	100.0	16.7
IF-P02	0.92	—*	88.5	100.0†	1.0	27.7	4	3	100.0	50.0
IF-P03	0.14	0.092	80.8	88.5	13.4	45.7	3	5	91.7	83.3
IF-P04	1.08	—*	100.0	100.0†	2.0	25.6	5	4	75.0	0.0‡
IF-P05	1.00	0.244	100.0	96.0	0.0	27.8	5	6	100.0	66.7
IF-P06	0.55	0.155	100.0	88.5	0.0	23.4	4	6	100.0	66.7
IF-P07	0.41	0.167	92.3	96.2	12.2	23.3	3	3	66.7	66.7
IF-P08	0.74	—*	96.2	100.0†	3.1	28.1	5	4	91.7	66.7
IF-P09	0.65	0.246	100.0	92.3	0.0	25.0	5	6	100.0	66.7
IF-P10	1.13	—*	100.0	100.0†	0.0	27.9	5	5	100.0	0.0‡
IF-P11	1.37	0.179	100.0	92.3	0.0	35.6	4	4	100.0	83.3

* Incomplete task in VR — excluded from speed comparison.
† Legibility computed over letters effectively written, not full alphabet.
‡ Score zeroed due to pen-lift detection threshold artifact.

Feasibility. All eleven participants engaged with the full protocol in both conditions without abandoning the session. Four participants did not complete the alphabet writing task (T01) in the VR condition, but proceeded normally through the remaining tasks; no participant interrupted the session or withdrew from the study. This is consistent with the findings of [5], who demonstrated that handwriting and sketching tasks in VR can be completed with low task load and high usability using a consumer-grade digitizing surface. The present study extends this evidence to a standardized handwriting assessment protocol, suggesting that VR-based administration is viable for this type of assessment.

Writing speed (T01). A systematic reduction in writing speed was observed in the VR condition across all seven participants with complete T01 data, with an average ratio of approximately 3.7×relative to the physical condition, ranging from 1.5× for IF-P03(the slowest writer on paper) to 7.7× for IF-P11 (the fastest).This finding aligns with [9], who document changes in handwriting dynamics when the writing surface is changed, and with the observations of [5], who note that handwriting performance in VR may not reach the same level as with a real pen, particularly during a first exposure to the virtual environment. The speed reduction is interpreted as a perceptual-motor adaptation effect stemming from the spatial decoupling between the writing surface and the visual feedback, where the participant writes on a physical digitizing tablet using a digital pen while observing the resulting strokes in the virtual environment through the HMD. This separation between the physical locus of motor execution and the virtual locus of visual output is a known source of performance adaptation in VR handwriting tasks [5].

Writing quality (T01 and T02). Despite the speed reduction, legibility in T01 remained relatively high in the VR condition, although among the seven participants who completed the task in both conditions,mean legibility was slightly lower in VR (92.3%) than on paper (95.6%).The four participants with incomplete tasks achieved 100% legibility over the letters they wrote, indicating that the interruptions were not caused by inability to produce legible handwriting. Similarly, the malformed letter rate in T02 was systematically higher in the VR condition (23.3%–45.7%) compared to physical (0%–13.4%), yet all participants produced evaluable handwriting output in both conditions.These results suggest that the VR environment affects handwriting dynamics more than it affects the fundamental motor ability to produce recognizable letterforms, which is consistent with the distinction between handwriting speed and handwriting quality described by [6].

Figure filling (T03). Task 03 showed the closest agreement between physical and VR conditions across the dataset. Four out of eleven participants obtained identical scores in both modalities, and no participant showed a difference greater than 2 points on the 6-point scale. This result is particularly relevant because T03 evaluates filling movement patterns (circular, vertical, and horizontal) which appear to transfer well to the VR tablet configuration. The absence of a timing constraint in this task may partially explain the closer agreement, as participants could adapt their pace without penalty.

Basic movement patterns (T04). Task 04 free movement results were affected by an overly restrictive pen-lift detection threshold,which produced zero scores for IF-P04 and IF-P10; this limitation is attributed to a system-level constraint in the current implementation and does not reflect participant handwriting ability. Excluding these two cases, free movement scores in VR ranged from 16.7% to 83.3%, with IF-P03, IF-P07, and IF-P11 achieving results within 17 percentage points of their physical scores. IF-P07 obtained an exact match (66.7% in both conditions), suggesting that the VR system is capable of capturing free movement patterns at a level comparable to the physical protocol once this artifact is corrected. Tracing performance was also affected by a separate technical limitation, described in Section VI,and is not compared across conditions in this paper due to the differing scales used to score it in each modality.

Limitations. This study presents several limitations that must be acknowledged. The pilot sample of N=11 participants does not allow for inferential statistical analysis, and the results

should be interpreted as exploratory. The identified technical artifacts in T04 — Y-threshold stroke separation and pen-lift detection — directly affected the comparability of results for a subset of participants and require correction before a full equivalence study can be conducted. In addition,all evaluation in the physical condition was performed by a single rater,following the manual scoring procedure defined in the original protocol [6]; although this preserves fidelity to the established method, it introduces a potential source of subjective bias absent in the automated VR evaluation. Future work should digitize the paper responses and score them with the same automated method applied in the digital condition, allowing both conditions to be evaluated under identical criteria. Finally, writing speed reduction in the VR condition may partially reflect the novelty effect of the virtual environment setup— in which the participant writes on a physical digitizing tablet while observing the output through the HMD — rather than a stable performance characteristic, suggesting that a familiarization session prior to data collection should be considered in future protocols.

VIII. Conclusion

This paper characterized differences between VR-based and paper-based handwriting assessment, applying an adapted version of the protocol proposed by Cardoso [6] in both conditions with eleven adult participants. Writing speed was substantially lower in VR than on paper, and letter malformation rates were consistently higher in VR,with non-overlapping ranges between conditions — differences that must be characterized and addressed before VR-based assessment can be treated as equivalent to its paper-based counterpart. Figure filling was the only task to show close agreement between modalities, suggesting that boundary- and direction-based movement patterns may transfer well to the proposed VR setup, while tasks that demand fine letter form production and writing speed are more strongly affected by the virtual environment.All participants engaged with the full protocol in both conditions,indicating that the setup is feasible for administering a structured handwriting assessment battery in VR, even though its outcomes do not yet match those obtained on paper.

Future work will address the identified technical limitations (the stroke separation and pen-lift detection artifacts), evaluate paper responses using the same automated scoring method applied to the digital condition, and conduct a study with a larger sample and a familiarization session, enabling formal agreement analysis to determine which measures, if any, are equivalent across conditions. Building on this foundation, the setup will support the development of a richer,gamified virtual environment aimed at increasing motivation and engagement with handwriting practice, particularly for children.

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